Lesson 5: Add a score

Pygame CE • S3 onwards • 40-50 min

Learning intention

Store game information in a variable and display text in the window.

Success criteria

- I can create and update a score variable.
- I can display text using a font.

Key ideas

- Text is drawn in two steps: **render** to an image, then **blit** it to the screen.
- f-strings such as `f"Score: {score}"` put variable values inside text.

Activities

1. Add a score variable and a font.
2. Increase the score on each collision.
3. Display the score, then change its colour and position.

Code

Key changes

```
score = 0
font = pygame.font.Font(None, 36)

# in the collision check:
score += 1

# after drawing objects:
score_text = font.render(f"Score: {score}", True, (255, 255, 255))
screen.blit(score_text, (10, 10))
```

Complete program

```
import pygame
import random

pygame.init()

WIDTH = 800
HEIGHT = 600
SIZE = 40

screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Score")
clock = pygame.time.Clock()

x = 380.0
y = 280.0
speed = 250
target = pygame.Rect(random.randint(0, WIDTH - SIZE),
                     random.randint(0, HEIGHT - SIZE), SIZE, SIZE)
score = 0
font = pygame.font.Font(None, 36)
running = True

while running:
    dt = clock.tick(60) / 1000
```

```
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    x += (keys[pygame.K_RIGHT] - keys[pygame.K_LEFT]) * speed * dt
    y += (keys[pygame.K_DOWN] - keys[pygame.K_UP]) * speed * dt
    x = max(0, min(x, WIDTH - SIZE))
    y = max(0, min(y, HEIGHT - SIZE))

    player_rect = pygame.Rect(int(x), int(y), SIZE, SIZE)
    if player_rect.colliderect(target):
        score += 1
        target.topleft = (random.randint(0, WIDTH - SIZE),
                          random.randint(0, HEIGHT - SIZE))

    screen.fill((30, 30, 40))
    pygame.draw.rect(screen, (240, 190, 60), target)
    pygame.draw.rect(screen, (80, 200, 120), player_rect)
    score_text = font.render(f"Score: {score}", True, (255, 255, 255))
    screen.blit(score_text, (10, 10))
    pygame.display.flip()

pygame.quit()
```

Checkpoint

The score goes up by exactly one for each target collected and stays visible.

Support and stretch

- **Support:** Give a worked text example to adapt.
- **Stretch:** Add a high score, or different targets worth different points.

Reflect

Which part of the program stores the score, and which part shows it?

> **Save your work**
> Save this version as `lesson5.py` before you start the next lesson.